

Taehyeon Kim

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PROFILE

Third-year Computer Science specialist at the University of Toronto (GPA 3.65/4.00), focused on machine learning and system architecture. Builds full-stack systems end to end, from production backend infrastructure and billing systems to ML pipelines, models, and fraud-detection experimentation, with deliberate attention to tradeoffs rather than just working code.

EDUCATION

University of Toronto

2024 – Present

B.Sc. Computer Science Specialist · GPA: 3.65 / 4.00

TECHNICAL SKILLS

Languages: Python, Java, C, TypeScript, Rust, Bash, SQL

Frameworks & Tools: FastAPI, Next.js, React, Axum/Tokio, PostgreSQL, Git, Linux, SQLAlchemy, Auth0, Tailwind CSS, Supabase

Cloud: AWS (IAM, VPC, EC2, S3, Lambda); pursuing AWS Certified Solutions Architect Associate

Concepts: System Design, OOP, Design Patterns, Data Structures & Algorithms, Scrum / Agile, REST APIs

AI / ML: scikit-learn, Gemini API, LLM integration, anomaly detection (Isolation Forest, Benford's Law, BiLSTM)

PROJECTS & HACKATHONS

MatiMath (AI Math Tutoring App)

2024 – Present

Long-term collaborative project · Team of 3

- Leading technical development of a Flutter tablet app that pairs an AI tutor character with step-based math exercises (fill-in-blank, matching, drawing, handwriting) and real-time TTS narration.
- Built Google Play Billing with server-side receipt verification through Firebase Cloud Functions as the sole writer of purchase entitlement, and cut the baked TTS asset pipeline from 476MB to 353MB.
- Wrote the privacy policy and account-deletion flow required for Google Play Families compliance.

chalk-talk (AI Blackboard Tutor Research)

2026 – Present

Solo research project

- Designing a closed-loop tutoring architecture where a vision-language model reads the canvas each step and decides the next spoken phrase or stroke, instead of generating a full script upfront.
- Built a working MVP that pre-generates a full lesson as JSON and animates it stroke-by-stroke with synced narration, used as the baseline and training-data source for the closed-loop research.

UTM Degree Explorer (Local-First Degree Planner)

May 2026

Solo project

- Built a full-stack degree planner over a fully scraped UTM Academic Calendar (2,346 courses, 164 programs), with a rule-based parser converting prerequisite text into structured ASTs at 86% coverage.
- Implemented live prerequisite validation and a React/Zustand planner UI with Supabase-backed multi-plan sync and guest mode.

Yosemite (AI Compliance & Fraud Detection Platform)

March 2026

GenAI Genesis 2026 · Team of 4

- Built the Rust (Axum) fraud-report API and integrated Google Gemini Vision and an LLM service for automated invoice and document analysis.
- Built the Next.js risk dashboard (risk overview, flagged transactions, protection score) and the PDF upload flow for the compliance platform.

Felix and Friends' ML Challenge (CSC311 Course Project)

Winter 2026

University of Toronto · Team of 4

- Ran feature-engineering ablations (StandardScaler normalization, TF-IDF vs. numeric-only) across three model families on a 3-class painting-classification task, showing TF-IDF removal dropped Bernoulli Naive Bayes accuracy from 93.5% to 76.7% and shaping the final unscaled, full-feature preprocessing pipeline.
- Analyzed and wrote up feature distributions (numeric, Likert, categorical) across 1,686 survey responses; the team's final neural network reached 94% test accuracy (macro F1 0.94).

Tango (Reinvented Job Matching Platform)

February 2026

DeerHacks V 2026 · Team of 3

- Implemented the core swipe/match feature (role-based discover queue, swipe UI, recruiter inbox) and integrated Auth0 authentication on a Next.js 15 frontend.
- Built a real-time chat feature with a matches sidebar and unread-message badges, and migrated image uploads from filesystem storage to PostgreSQL.

WORK EXPERIENCE

Crew Member

June 2026 – September 2026

Chipotle Mexican Grill

- Ran a high-volume, fast-paced kitchen line, building strong time management and prioritization skills while delivering friendly, consistent customer service during peak rush periods.

Math, Physics & Chemistry Tutor

September 2024 – 2026

Self-employed

- Provided personalized academic support to high school students; adapted explanations to individual learning gaps rather than following fixed curricula.

Customer Service Associate

Summer 2025

Korean Supermarket

- Developed conflict resolution and negotiation skills in a high-volume customer-facing environment.

LEADERSHIP

Finance Associate

2026 – Present

UTM Mathematical & Computational Sciences Society (MCSS)

- Volunteer on the student executive team representing all full-time Mathematical & Computational Sciences majors at UTM, supporting audit planning and financial reporting.
- Help run department events and day-to-day club operations alongside the team, in a volunteer capacity.

VP Finance

2025 – 2026

UTM Salsa & Bachata Club

- Managed the club's annual budget and financial operations for a year, overseeing event expenses, sponsorships, and financial reporting.

ADDITIONAL

Languages: Fluent in English and Korean**Presentations:** Weekly presentation experience over three years; comfortable communicating technical concepts to varied audiences.